

STUDY QUESTIONS 4

1. **Corner Point Solution Method-** Consider given linear programming model:

$$\begin{aligned}
 &\text{minimize} && z = 6x + 5y \\
 &\text{subject to} && \\
 &&& 3x - y \leq 6 && \text{(C1)} \\
 &&& -x + 4y \geq 12 && \text{(C2)} \\
 &&& x + y \leq 10 && \text{(C3)} \\
 &&& x \geq 2 && \text{(C4)} \\
 &&& x, y \geq 0 && \text{(C5)}
 \end{aligned}$$

- a) Plot the graph of this model and show the feasible region and corner points.
- b) How many corner point solutions does this model have? Classify each corner point solution by their feasibility.
- c) Solve the linear programming model using corner point solution method.
- d) Consider the given model without constraint C3. How does the feasible region change? Does the optimal solution remain the same or change? If the problem were a maximization problem, how would the removal of C3 affect the solution?

2. **Graphical Solution Method-** Consider given linear programming model:

$$\begin{aligned}
 &\text{maximize} && z = 5x + y \\
 &\text{subject to} && \\
 &&& 2x + 3y \leq 12 && \text{(C1)} \\
 &&& x - y \leq 2 && \text{(C2)} \\
 &&& 2y \leq 3 && \text{(C3)} \\
 &&& 2x \geq 3 && \text{(C4)} \\
 &&& x, y \geq 0 && \text{(C5)}
 \end{aligned}$$

- a) Determine if the linear programming model is feasible or not. If feasible, solve the model graphically by showing the objective function contour at optimal solution point. If it is not feasible, explain why it is not feasible.
- b) Consider the case where we multiplied constraint C1 by some scalar $c \in R$ and added that constraint to our linear programming model as constraint C6. Do the nature of model, optimal value and optimal solution change? Does the upper bound for number of corner point solutions change?

3. **An Interesting Model-** Consider given linear programming model:

$$\begin{aligned}
 &\text{minimize} && z = 6x + 7y \\
 &\text{subject to} && \\
 &&& 4x + 5y \leq 40 && \text{(C1)} \\
 &&& -x + 4y \leq 12 && \text{(C2)} \\
 &&& x \geq 6 && \text{(C3)} \\
 &&& x, y \geq 0 && \text{(C4)}
 \end{aligned}$$

- a) Determine if the linear programming model is feasible or not. If feasible, solve the model graphically by showing the objective function contour at optimal solution point. If it is not feasible, explain why it is not feasible.
- b) Consider the same model with constraint $C5 : 2x + y \geq 30$ added. Determine the upper bound for corner point solutions. If the linear programming model is feasible, solve the model with corner points method. If it is not feasible, explain why it is not feasible.
- c) Consider the following statement: “For any linear programming model that has unbounded feasible region, the model results in an unbounded objective function?” Is it true? Why or why not?

4. **A single-server coffee shop system-** A small coffee shop opens at time $t = 0$. There is only one barista working in the shop, so customers are served one at a time. The service discipline is FIFO (first-come, first-served).

Four customers arrive during the observation period. The first customer arrives as soon as the coffee shop opens and requires 4 minutes of service. The second customer arrives 3 minutes after opening and requires 1 minute of service. The third customer arrives 6 minutes after opening and requires 4 minutes of service. Finally, the fourth customer arrives 8 minutes after opening and requires 2 minutes of service.

Assume that the barista starts serving a customer immediately if the barista is idle. If the barista is busy, the arriving customer waits in the queue until all earlier customers have been served.

- a) Calculate the Service Start Time, Departure Time, Waiting Time in Queue, and Average Waiting Time for each customer.
 - b) Calculate the Server Utilization over the period from the opening of the coffee shop until the last customer departs.
5. **Campus print kiosk system-** A small print kiosk on a university campus opens at noon. Assume that noon is minute 0. The kiosk has only one employee, so students are served one at a time. The service discipline is FIFO (first-come, first-served). No student leaves the queue once they join.

Although the kiosk opens at noon, the employee needs 2 minutes to prepare the printer and the payment system. Therefore, the employee can only begin serving students starting from minute 2. Students who arrive before service begins wait in line until the employee is ready.

The inter-arrival times and service times for the first five students are given below. The first inter-arrival time is measured from the opening time of the kiosk. All times are in minutes.

Student	Inter-arrival time (min)	Service time (min)
1	1	3
2	4	2
3	1	5
4	2	1
5	6	4

- a) For each of the five students, determine: (i) arrival time, (ii) service start time, (iii) service completion (departure) time, and (iv) waiting time in queue.
 - b) Using your results from part (a), calculate: (i) the average waiting time across all students, (ii) the maximum number of students waiting in queue at any point in time, and (iii) the server utilization during the period from the opening of the kiosk until the last student departs.
6. **Advertising and Sales-** A store manager wants to understand how advertising spending affects product sales. In previous promotional campaigns, the manager recorded the amount of money spent on advertising and the corresponding number of units sold. The goal is to use this past data to build a simple prediction model for future sales.

In this model, let x represent the advertising spending in dollars, and let y represent the number of units sold. The past observations are given below:

Advertising(x)	Sales (Units) (y)
\$100	1000
\$200	1500
\$400	3000
\$300	2500

Assume that the relationship between advertising spending and sales can be approximated by a simple linear regression model: $\hat{y} = \beta_0 + \beta_1 x$

- Using the data above, manually calculate the slope (β_1) and the intercept (β_0) of the regression line. Write regression model.
- Use the fitted regression model to estimate the expected number of units sold if the manager spends \$150 on advertising.

7. **A real estate company-** A real estate company wants to predict house prices based on past sales data. Three employees in the company have different opinions about which factor is the most useful for predicting the selling price (y) of a house.

The first employee believes that house prices mainly depend on the size (x_s) of the house. The second employee believes that the age (x_a) of the house is the most important factor. The third employee believes that whether the house is located in the city center (x_c) is the key factor.

The company manager asks each employee to build a separate simple linear regression model using only one explanatory variable.

In the data set, price represents the selling price of the house in thousand dollars. Size represents the size of the house in thousand square feet. Age represents the age of the house in years. The variable center is a binary variable: it is equal to 1 if the house is located in the city center and 0 otherwise.

The data collected from five recently sold houses are given below:

House	Size (sq ft *1000)	Age (years)	Center	Price (\$ *1000)
1	1	8	0	180
2	2	12	1	280
3	3	30	0	170
4	4	25	0	210
5	5	20	1	430

- For each employee, write the estimated regression model ($\hat{y} = \beta_0 + \beta_1 x$) and calculate the coefficient of determination (R^2) for each model.
- Compare the three simple linear regression models. Which employee's model would you prefer for prediction using this data? Briefly explain your reasoning.
- A real estate analyst joins the meeting and suggests the following two multiple linear regression models, fitted using the same data. Which model would you prefer? Briefly explain your reasoning.

• **Model SA:**

$$\hat{y}_{sa} = 206,480 + 81.16(x_s) - 10,314(x_a) \quad (R_{sa}^2 = 0.842)$$

• **Model SC:**

$$\hat{y}_{sc} = 110,545 + 28.55(x_s) + 144,545(x_c) \quad (R_{sc}^2 = 0.899)$$

- Can we claim that a new model using all three variables (Size, Age, and Center) will certainly achieve an in-sample R^2 value that is not worse than Models A and B? Briefly explain your answer.

8. **Supplier selection-** A small electronics manufacturer needs 500 units of a component for next month's production. The company can purchase the component from three different suppliers. Each supplier has a unit purchasing cost, a fixed contracting cost, and a maximum supply capacity. If a supplier is selected, the company must pay the fixed contracting cost for that supplier.

In addition, due to supplier contract rules, if a supplier is selected, at least 100 units must be purchased from that supplier.

The supplier information is given below:

Supplier	Unit cost	Fixed cost	Capacity
A	5	700	300
B	6	300	250
C	4	900	400

The company wants to decide which suppliers to select and how many units to purchase from each selected supplier so that the total cost is minimized.

- Define the decision variables for this problem.
 - Write: (i) The objective function that minimizes the total cost, (ii) The demand satisfaction constraint, (iii) The capacity and "only buy if selected" linking constraints, (iv) The minimum purchase constraints stating that if a supplier is selected, at least 100 units must be purchased from that supplier, (v) The variable domain restrictions.
9. **Distribution center opening problem-** A retail company is planning to open distribution centers (DCs) to serve four stores in a region. Each potential DC has a fixed opening cost and a limited capacity. Each store has a known demand, and shipping one unit of product from a DC to a store incurs a unit transportation cost.

The company must decide which DCs to open and how much to ship from each open DC to each store. All store demands must be satisfied, and the capacity of each opened DC cannot be exceeded. The objective is to minimize the total cost, including both fixed opening costs and transportation costs.

The fixed opening costs and capacities of the DCs are given below:

DC	Fixed opening cost	Capacity
1	650	70
2	500	60
3	300	45

The store demands are:

Store	Demand
A	30
B	35
C	25
D	25

The unit transportation costs are:

	Store A	Store B	Store C	Store D
DC 1	4	7	6	5
DC 2	6	4	5	8
DC 3	8	6	4	3

- Define the decision variables for this problem.
- Write: (i) The objective function that minimizes the total cost, (ii) The demand satisfaction constraints for all stores, (iii) The capacity constraints for all DCs, (iv) The linking constraints that prevent shipments from a DC unless that DC is opened.

- c) Based only on capacity, determine which pairs of DCs would be feasible if exactly two DCs were opened. Briefly explain your answer.
10. **Delivery route planning-** A courier company needs to deliver packages to six locations: A, B, C, D, E, and F. The courier starts from location A, visits each other location exactly once, and returns to A at the end of the route.

The travel distances between locations are given in the table below. The distance matrix is symmetric.

From / To	A	B	C	D	E	F
A	0	11	16	22	13	19
B	11	0	10	15	9	12
C	16	10	0	8	11	7
D	22	15	8	0	12	10
E	13	9	11	12	0	14
F	19	12	7	10	14	0

- a) Apply the nearest neighbor heuristic to construct an initial TSP tour.
- b) Write the complete initial tour and calculate its total distance.
- c) Apply the following 2-opt move to the initial tour: remove edges $B - E$ and $D - A$, and add edges $B - D$ and $E - A$. Calculate the old edge cost and the new edge cost.
- d) Does this 2-opt move improve the tour? If yes, write the improved tour and its new total distance. If not, explain why the move should be rejected.